

TrailMark Project OS

Version 1.0

Chapter 6 – Feature Architecture

Purpose

Feature Architecture defines every capability TrailMark will eventually offer. Each feature exists to deepen a visitor's connection with a National Park and should fit naturally into the overall ecosystem.

Core Platform

Website, iPhone app, user accounts, personal archive, badge collection, search, offline sync, cloud backup, notifications, accessibility, and settings form the platform foundation.

Exploration Features

Interactive trail maps, GPS location awareness, Digital Ranger, hidden viewpoints, nearby recommendations, estimated hiking times, scenic drives, and itinerary planning.

Learning Features

Hidden history, geology explanations, wildlife education, Indigenous perspectives, ecosystems, astronomy, conservation, ranger stories, and 'Did you know?' discoveries.

Photography System

Photography Coach, golden hour planning, Milky Way guidance, composition tips, lens suggestions, reflection finder, waterfall conditions, seasonal color forecasts, and challenge missions.

Safety System

Weather-aware suggestions, wildlife alerts, emergency information, accessibility guidance, offline maps, camping preparation, gear recommendations, and daylight warnings such as '30 minutes until sunset.'

Journal & Collection

Field journal entries, favorite places, trip timelines, personal notes, photo collections, earned badges, collectible posters, park completion history, and memory preservation.

Family Experiences

Kids Mode, Junior Ranger-inspired activities, educational games, scavenger hunts, quizzes, audio adventures, and age-appropriate learning paths.

Community Layer

Optional community experiences focused on stewardship and learning: shared journals, photography inspiration, park challenges, and conservation participation rather than follower counts.

Commerce

Premium posters, badge sets, books, field journals, apparel, limited editions, gift collections, and future memberships should all connect back to the same illustrated archive.

AI Systems

The Digital Ranger becomes the intelligence layer that connects every feature. It understands location, weather, season, visitor interests, accessibility needs, available time, photography goals, and previous visits to make thoughtful recommendations.

Architecture Principles

Every feature must answer three questions: Does it help someone discover something meaningful? Does it deepen their appreciation for the park? Does it strengthen the lifelong TrailMark archive? If not, it does not belong.